

Edam Hamza

+216 93 060 715 — edam.hamza@supcom.tn — Edam Hamza — EdamH

Summary — I am a final-year ICT Engineering student at **SUP'COM**, with a strong interest in **Generative AI** and **Data Engineering**. My focus is on gaining expertise in these fields by utilizing **DevOps tools**, and I am actively seeking opportunities to deepen my knowledge and skills in this area.

Education

Higher School of Communication of Tunis (SUP'COM) **September 2022 - Present**
Relevant Coursework: Big data, cloud computing, database management, Data Engineering and networking

Preparatory Institute for Engineering Studies of Sfax **2020 - 2022**
National Engineering School Entrance Exams : Math and Physics - Rank: 89/1831

Experience

EY - Tutored Project **September 2024 – Present**
Optimization of Data Structures Testing Using Generative AI **Tunis**
Technologies: Machine Learning, NLP, Neural Networks, Python, SQL, APIs, DBMS

- Automated **data structure testing** using **AI**, applying **machine learning** for **schema optimization**.
- Leveraged **generative AI** to develop **validation models**.
- Integrated **AI** with **DBMS** via **APIs**, enabling seamless operation, scalability, and enhanced performance through **deep learning** and **NLP-driven analysis**.

HTWK Leipzig - Summer Internship **July 2024 – September 2024**
Development of an Intelligent Camera System for the Detection & Automatic Centering of Electric Poles **Leipzig**
Technologies : OpenCV, Python, ArduinoIDE, Flutter, Bluetooth LE, UART, Altium, Blender

- Coded an algorithm for detecting **markers** in images and their offset using **Canny edge detection**.
- Built an **embedded controller** to correct camera position on all three axes using **servo motors** and a **printed gimbal**.
- Programmed an **Android app** to display images and parameters, with an option for modification through **BLE**.

ACTEOL CRM - Summer Internship **June 2023 – August 2023**
AI-Driven CRM Insights **Sfax**
Technologies: Vite.js, .NET MVC, Microsoft SQL Server, OpenAI API, LangChain, LLMs, GraphAPI

- Developed a full-stack application using **Vite.js**, **.NET MVC**, and **Microsoft SQL Server** to analyze data from **Meta**.
- Integrated the **OpenAI API** and **LangChain** to train **GPT-3** on embeddings generated from clients' Facebook pages.

Projects

Real-Time Flight into Data Pipeline **July 2024**
Technologies: Kafka Spark Elasticsearch Kibana Docker GitHub Actions Azure

- Developed real-time flight tracking with Kafka, Spark, Elasticsearch, and Kibana; automated CI/CD with GitHub Actions; deployed on Azure.
- Implemented unit tests to ensure functionality.

Scalable MERN Slack-Clone with Docker Swarm & Jenkins CI/CD **January 2024 – May 2024**
Technologies: MERN Stack, Docker Swarm, Jenkins, Nginx, Prometheus, Grafana, Jest.js, Microsoft Azure

- Built a scalable Slack-clone using MERN, containerized with Docker Swarm for automatic scaling.
- Automated CI/CD pipeline with Jenkins for testing, integration, and deployment across multiple environments.
- Integrated Jest.js for unit testing, with smoke and integration tests for reliability.
- Enhanced security using Nginx reverse proxy, preventing XSS, HTTP pollution, and DDoS attacks.
- Deployed on Microsoft Azure, with monitoring via Prometheus, cAdvisor, and Grafana.

Skills

Data Engineering Kafka, Spark, Kibana, Grafana	Cloud & DevOps Azure, GitHub Actions, Jenkins, Docker, AWS
Generative AI LangChain	Database Elasticsearch, MongoDB, MySQL, PostgreSQL
Languages Python, JavaScript, Java, C++, C#, Dart	OS Ubuntu, Windows

Volunteer Experience & Organisations

IEEE Sup'Com Student Branch **May 2023-June 2024**
Chair of The Membership Committee

IndabaX Tunisia 2024 **January 2024-May 2024**
Supervisor